

Team Worksheet — Lab 27

Statistical Process Control and Chart Selection — In Depth

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-27-spc-control-charts.xlsx`.

3. What we produced

Deliverable: A correctly selected and constructed control chart for your capstone Y with applied run rules.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap SPC purpose in full: distinguish signal from noise over time, so drift is caught before it becomes a defect.

Step 2. Recap chart anatomy: centre line, upper and lower control limits at ± 3 sigma, and why 3 sigma balances false alarms against missed signals.

Step 3. Establish the critical distinction: CONTROL limits come from the process's own voice (the data); SPECIFICATION limits come from the customer. They are unrelated and must never be drawn on the same axis as if comparable.

Step 4. Learn the chart selection logic in full. Continuous data: subgroup size 1 = I-MR; 2-8 = Xbar-R; 9+ = Xbar-S.

Step 5. Learn the attribute branch: defectives with constant sample size = np; varying = p. Defects with constant opportunity = c; varying = u.

Step 6. Select the correct chart for your capstone Y and write down the justification. Selecting the wrong chart produces wrong limits and false confidence.

Step 7. Collect at least 25 subgroups of baseline data — fewer gives control limits too unstable to be trustworthy.

Step 8. Calculate the control limits using the correct constants (A2, D3, D4 for Xbar-R; equivalent constants for other charts).

Step 9. Plot the chart and apply the Western Electric run rules in full: 1 point beyond 3 sigma; 2 of 3 beyond 2 sigma; 4 of 5 beyond 1 sigma; 8 consecutive on one side; trends; and hugging the centre line.

Step 10. Investigate every signal. If a special cause is found and removed, recalculate the limits excluding that point — but only with a documented assignable cause.

Step 11. Confirm the process is stable before proceeding. An unstable process has no meaningful capability and cannot be controlled by a control plan alone.

7. Self-check

Your chart selection is justified against the selection logic, limits are computed from 25+ subgroups with the correct constants, and all run rules are applied with every signal investigated.

Complete? ■ Yes ■ Not yet

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